American Computer Science League

2025-2026 ● Contest 3: Short Problems Solutions ● Junior Division

1. Boolean Algebra

$$A(\overline{B + C}) + \overline{ABC} = A\,\overline{B}\,\overline{C} + \overline{A} + \overline{B} + \overline{C}$$
$$= \overline{B}(A\,\overline{C} + 1) + \overline{A} + \overline{C}$$
$$= \overline{A} + \overline{B} + \overline{C}$$

This is FALSE when each term is 0, so $\overline{A} = 0, \overline{B} = 0, \overline{C} = 0$.
The only ordered triple would be (1,1,1).

D. (1,1,1)

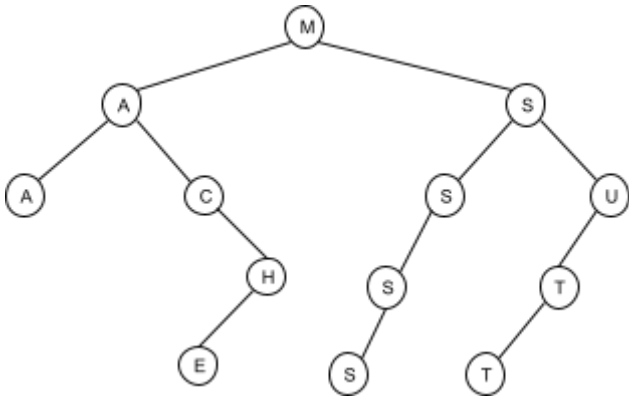
2. Boolean Algebra

$$\overline{AB}\,(\overline{A + C}) + \overline{C}$$
$$= (\overline{A} + \overline{B})\,(\overline{A} * \overline{C}) + \overline{C}$$
$$= \overline{A} * \overline{C} + \overline{A} * \overline{B} * \overline{C} + \overline{C}$$
$$= (\overline{A} + \overline{A} * \overline{B} + 1)\,\overline{C}$$
$$= 1 * \overline{C}$$
$$= \overline{C}$$

C. $\overline{C}$

3. Data Structures

The binary search tree for MASSACHUSETTS is:



One child nodes are: C, H, S, S, T, U. So there are 6 of them.

D. 6

4. Data Structures

The stack is constructed using LIFO as follows:
B, BO, B, BS, B, BT, BTO, BT, B, BN, B
Therefore, the next item popped is B.

E. B

5. What Does This Program Do? (Arrays)

arr =

1	4	9	16	25

```
num = 30
count = -1
while count != 3
    count = 0
    for y = 0 to 4
        if num%arr(y) == 0 then
            count = count + 1
        end if
    next y
    if count < 3 then
        num = num + 1
    end if
end while
output num
```

32 is the smallest number that has 3 proper factors that are perfect squares since its proper factors are 1, 2, 4, 8, and 16.

A. 32

6. What Does This Program Do? (Arrays)

```
p = 1
for x = 0 to 5
    m = 2
    for y = 0 to 1
        if arr(y,x)<arr(m,x) then
            m = y
        end if
    next y
    if arr(m,x) < 5 then
        p = p * arr(m,x)
    end if
next x
output p
```

This program finds the minimum value in each column. It finds the product of all minimum numbers that are less than 5.They are 1, 2, 3, and 4 so the product is 24.

D. 24